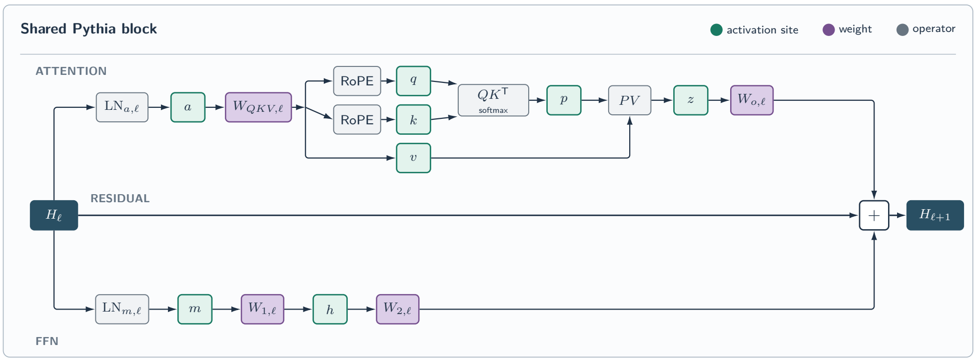
Status Report Number 1

Thiago Mattar
thiagocmattar@gmail.com

31 August 2026

1 Architecture and intervention ladder

Figure 1 locates the activation sites and organizes the progressive experimental program.



Sparsification ladder

Non-linearity: ● GeLU ● ReLU ● $G_{+,\kappa}$ ● $G_{\pm,\kappa}$ ● none

Labels		Intervention								$R^{\max}_{\text{model}}$ at $T = 2{,}048$		
Step	ID	h	m	a	z	q	k	v	Pressure	14M	70M	410M
1	A0	GeLU	none	none	none	none	none	none	none	0.0%	0.0%	0.0%
2	A1-H	ReLU	none	none	none	none	none	none	none	4.3%	12.4%	24.9%
3	A1-H-L1	ReLU	none	none	none	none	none	none	L1N	4.3%	12.4%	24.9%
4	A1-H-OL1	ReLU	none	none	none	none	none	none	OL1	4.3%	12.4%	24.9%
5	A4	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	none	none	none	none	12.8%	37.1%	74.8%
6	A4-OL1	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	none	none	none	OL1	12.8%	37.1%	74.8%
7	A7	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{\pm,\kappa}$	$G_{\pm,\kappa}$	$G_{\pm,\kappa}$	none	30.0%	49.4%	87.2%
8	A7-OL1	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{+,\kappa}$	$G_{\pm,\kappa}$	$G_{\pm,\kappa}$	$G_{\pm,\kappa}$	OL1	30.0%	49.4%	87.2%

Figure 1: Shared Pythia block and sparsification intervention ladder. A4 and A7 denote operational topologies A4-Z and A7-Z-POST.

2 Experiment control

2.1 Scope dashboard

Pilot. Local execution with 581 optimizer boundaries, global batch 64, and 76,152,832 training input tokens per condition.

Paper-scale full pass. RunPod A100 execution with 712 optimizer boundaries, global batch 1,024, and 1,493,172,224 training input tokens per condition. Conditions run concurrently across independent GPUs.

Table 1: Experiment scope at the report cutoff.

Step	Conditions/model	Pythia-14M	70M	410M
A0	1	Full pass complete: Run 004	0/1	0/1
A1-H	1	Full pass complete: Run 004	0/1	0/1
A1-H-L1	4	4/4 full pass: Run 004	0/4	0/4
A1-H-OL1	4	4/4 full pass: Run 009	0/4	0/4
A4	5	5/5 full pass: Run 011	0/5	0/5
A4-OL1	5	5/5 full pass: Run 015	0/5	0/5
A7	5	5/5 full pass: Run 013	0/5	0/5
A7-OL1	5	5/5 full pass: Run 014	0/5	0/5

2.2 Paper-scale contract and parameter set

The scale plan now adopts the size-specific learning rates in the standard Pythia recipe: peak/minimum $10^{-3}/10^{-4}$ for 14M and 70M, and $3 \times 10^{-4}/3 \times 10^{-5}$ for 410M. The study applies each activation intervention within this established recipe.

Table 2: Shared paper-scale execution contract.

Field	Specification
Model construction	Random initialization from pinned Pythia architecture; model seed 1234
Data	Pinned MiniPile cache; data-order seed 1234; sequence length 2,048
Training horizon	712 optimizer boundaries; 1,493,172,224 input tokens per condition
Global batch	1,024 sequences; hardware-specific microbatch and accumulation preserve the product
Optimizer	AdamW, betas $(0.9, 0.95)$, epsilon 10^{-8}, weight decay 0.1, global clipping 1.0
Schedule	1% linear warmup; cosine decay to 10% of the size-specific peak learning rate
Precision	Dynamic FP16 autocast; FP32 parameters and optimizer state
Validation	500 documents; 338 complete blocks; 692,224 input tokens; 1,444-token tail recorded
Retention	Activation counts and moments, weight norms, logical counts, pressure-gradient metrics, final checkpoint

Table 3: Aggregated intervention parameter set. Threshold equality survives.

Family	Gate and pressure	Parameter grid
A0	Stock GeLU at `h`	Singleton
A1-H	ReLU at `h`	Singleton
A1-H-L1	ReLU; naive L1 at `h`	$\lambda \in \{0.05, 0.1, 0.5, 1.0\}$
A1-H-OL1	ReLU; OL1 at `h`	Same λ grid; budget 1.0
A4	One-sided threshold at `a,m,h,z`	$\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$
A4-OL1	A4 gates; OL1 at four sites	Same κ grid; $\lambda = 1$; budget 1.0
A7	A4 plus symmetric `q_post,k_post,v` gates	Same κ grid
A7-OL1	A7 gates; OL1 at seven sites	Same κ grid; $\lambda = 1$; budget 1.0

3 RunPod compute

Table 4 reports elapsed wall-clock time for each condition's scientific process. Latest recorded RunPod cost across the six main-ladder runs is at least \$88.39; the retained 100 GB network volume remains separate at \$7/month.

Table 4: RunPod execution record. Wall-clock is elapsed scientific-process time per completed condition.

Run	Compute	Wall-clock	Cost
004	5× A100 80 GB; six conditions	63–108 min	$21.52
009	4× A100 80 GB; four conditions	86–105 min	$12.27
011	5× A100 80 GB; five conditions	59–104 min	$11.47
015	5× A100 80 GB; five conditions	94–101 min	$13.17*
013	5× A100 80 GB; five conditions	82–89 min	$15.83
014	5× A100 80 GB; five conditions	74–115 min	$14.13
Total	**Latest recorded spend**	—	**≥$88.39**

*Run 015's $13.17 is provisional because billing posting lagged teardown. Deployment required an 80 GB memory class, cache staging, detached execution, resumable artifact transfer, hash verification, and explicit teardown. The final Run 015 audit found zero Pods; the pre-existing retained volume was unchanged. Historical Run 012 compute and the combined spend including it are retained in Appendix A.

4 Results

The analysis adds one intervention at each stage and maps the resulting validation-loss–logical-compute frontier. Only completed comparisons appear as result subsections.

All site columns report exact-zero percentages; `q` and `k` denote the post-RoPE tensors. "n.m." denotes a site that was not measured in the source run.

Table 5: Progressive analysis status.

Stage	Comparison	Evidence status
1	A0 → A1-H	Full-pass endpoints complete
Control	A0/A1-H → uniform post-hoc clipping	Full-pass Analysis 005 complete
2	A1-H → A1-H-L1	Full-pass dose response complete
3	A1-H-L1 → A1-H-OL1	Matched full-pass Analysis 003 complete
4	A1-H → A4	Matched full-pass Analysis 004 complete
5	A4 → A4-OL1	Corrected Run 015 in Analysis 009; descriptive; F001 discarded
6	A4 → A7	Matched full-pass Analysis 008; tentative F002
7	A7 → A7-OL1	Five matched full-pass pairs in Analysis 008; descriptive

4.1 Type of non-linearity: A0 to A1-H

ReLU creates a 63.448% exact-zero rate at `h` and moves R_{model} to 2.714%.

Table 6: Pythia-14M full-pass controls from Run 004.

Condition	Loss	R_{model} (%)	Exact-zero (%)						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
A0: GeLU	5.209	< 0.001	< 0.001	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
A1-H: ReLU	5.270	2.714	63.448	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

The evaluation-only uniform TEAL-style control thresholds `a`, `m`, `h`, and `z` separately in every block; it does not threshold `q`, `k`, or `v`. For one common target p, values satisfying $|x| \leq t$ are zeroed without retraining. Table 7 is the only result table that reports paired loss deltas.

Table 7: Selected uniform TEAL-style post-hoc endpoints from Analysis 005, Observation O001. Deltas are paired to the independently evaluated $p = 0$ row for the same checkpoint.

	GeLU control			ReLU control		
p	**Loss**	**Δloss**	R_{model} (%)	**Loss**	**Δloss**	R_{model} (%)
0.0	5.208594	0.000000	< 0.001	5.269633	0.000000	2.714133
0.1	5.214453	0.005859	1.278706	5.274073	0.004439	3.568403
0.2	5.250270	0.041676	2.554786	5.307940	0.038307	4.418365
0.3	5.361901	0.153307	3.831641	5.405056	0.135422	5.271871
0.4	5.605384	0.396790	5.125544	5.602965	0.333331	6.129325
0.5	6.077731	0.869137	6.441923	5.976274	0.706641	6.994451

4.2 Naive L1 pressure: A1-H to A1-H-L1

Increasing λ raises exact-zero mass and R_{model} across the four full-pass endpoints.

Table 8: Run 004 full-pass naive-L1 dose response.

λ	**Loss**	R_{model} (%)	**Exact-zero (%)**						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.05	5.206	3.142	73.441	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.10	5.165	3.336	77.993	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.50	5.113	3.788	88.542	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
1.00	5.102	3.949	92.323	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

4.3 Orthogonalization: A1-H-L1 to A1-H-OL1

The matched full-pass endpoints occupy a narrow common region in validation loss, R_{model}, and `h` sparsity.

Table 9: Absolute naive-L1 and OL1 endpoints from Analysis 003.

λ	**Method**	**Loss**	R_{model} (%)	**Exact-zero (%)**						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.05	L1	5.206	3.142	73.441	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.198	3.145	73.526	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.10	L1	5.165	3.336	77.993	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.159	3.339	78.046	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.50	L1	5.113	3.788	88.542	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.110	3.768	88.084	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
1.00	L1	5.102	3.949	92.323	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.121	3.938	92.067	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

Figure 2 isolates the OL1 Adam-relative geometry before and after conflict projection.

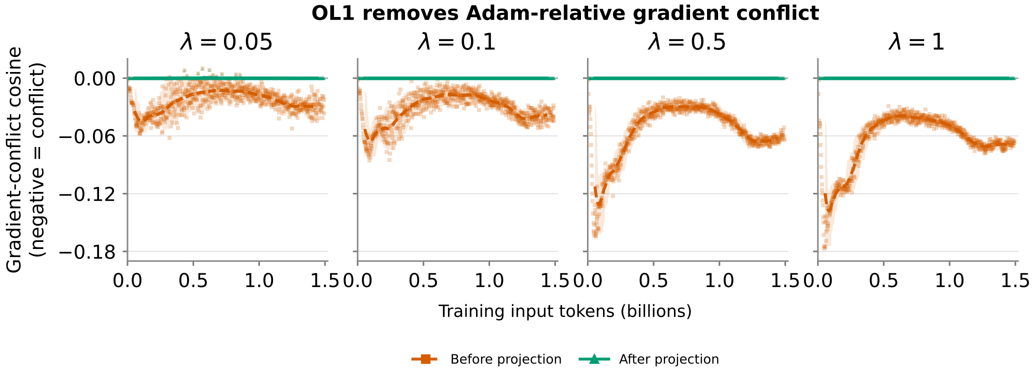


Figure 2: OL1 Adam-relative interaction before and after conflict projection. Each panel contains 712 optimizer boundaries for one λ; lines are centered 51-boundary means and bands span p05–p95. Source: Analysis 003, Observation O004.

4.4 Architecture-wide thresholding and OL1: A4 to A4-OL1

A4 applies one-sided thresholds at `a`, `m`, `h`, and `z`. The corrected A4-OL1 objective in Run 015 adds post-gate OL1 pressure at all four sites with $\lambda = 1$ and trust budget 1.0. It improves both validation loss and R_{model} over matched A4 at $\kappa = 0$ and 0.01. At $\kappa = 0.05$, 0.1, and 0.5, it adds 2.2506, 2.4413, and 2.4979 percentage points of R_{model} for validation-loss costs of 0.055693, 0.128681, and 0.378307. This one-seed result remains descriptive; discarded Finding F001 is not restored.

Table 10: Absolute A4 and corrected four-site A4-OL1 full-pass endpoints from Analysis 009, Observation O001. Every percentage is recomputed after pooling integer counts over all six layers and 338 complete validation blocks.

κ	Method	Loss	R_{model} (%)	Exact-zero (%)						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	A4	5.470	7.212	68.765	49.779	48.560	54.525	< 0.001	< 0.001	< 0.001
	A4-OL1	5.458	7.810	74.205	39.296	68.891	69.625	< 0.001	< 0.001	< 0.001
0.01	A4	5.466	7.414	71.181	50.397	49.095	59.643	< 0.001	< 0.001	< 0.001
	A4-OL1	5.458	8.587	80.833	43.464	73.491	85.259	< 0.001	< 0.001	< 0.001
0.05	A4	5.434	8.206	81.340	52.712	51.155	77.641	< 0.001	< 0.001	< 0.001
	A4-OL1	5.490	10.456	93.275	60.439	88.505	97.392	< 0.001	< 0.001	< 0.001
0.10	A4	5.420	8.953	91.853	55.200	53.244	89.234	< 0.001	< 0.001	< 0.001
	A4-OL1	5.548	11.394	97.149	75.912	91.310	99.283	< 0.001	< 0.001	< 0.001
0.50	A4	5.660	10.216	99.876	66.386	63.434	99.881	< 0.001	< 0.001	< 0.001
	A4-OL1	6.038	12.713	99.940	96.766	98.906	99.976	< 0.001	< 0.001	0.661

4.5 Attention-operand thresholding and OL1: A4 to A7 to A7-OL1

A7 adds symmetric thresholds at post-RoPE `q`, `k`, and `v` to A4's one-sided `a`, `m`, `h`, and `z` gates; the matched A4/A7 result supports tentative Finding F002. A7-OL1 adds post-gate orthogonal L1 pressure at all seven active sites with $\lambda = 1$ and trust budget 1. At matched $\kappa = 0.1$, A7-OL1 adds 1.3717 percentage points of R_{model} for +0.000816 validation loss; at $\kappa = 0.5$, it adds 12.0959 points for +0.126466 loss. The zero-threshold row regresses, so the A7/A7-OL1 effect is threshold-dependent and remains descriptive.

Table 11: Matched A7 and A7-OL1 full-pass endpoints from Analysis 008, Observation O001. Every percentage is recomputed after pooling integer counts over all six layers and 338 complete validation blocks. `out` is the ungated post-W_o `attention_output`.

κ	Variant	Loss	R_{model} (%)	Exact-zero mass (%)							
				`h`	`m`	`a`	`z`	`q`	`k`	`v`	`out`
0.00	A7	5.468401	7.217651	68.689	49.790	48.590	55.221	< 0.001	< 0.001	< 0.001	< 0.001
	A7-OL1	5.480184	7.054229	68.277	49.503	48.583	42.761	< 0.001	< 0.001	< 0.001	< 0.001
0.01	A7	5.458822	7.617649	71.275	50.330	49.064	59.041	0.399	0.407	1.654	< 0.001
	A7-OL1	5.475797	7.723393	71.050	50.103	49.196	48.933	1.373	1.496	2.338	< 0.001
0.05	A7	5.437888	9.126924	81.298	52.681	51.036	78.018	1.840	1.716	7.312	< 0.001
	A7-OL1	5.462811	9.863032	81.052	52.241	51.036	72.067	5.427	5.817	10.526	< 0.001
0.10	A7	5.428681	10.425018	91.353	55.075	53.052	90.052	3.150	3.227	11.355	< 0.001
	A7-OL1	5.429497	11.796760	91.607	54.874	53.104	88.896	8.227	8.544	18.819	< 0.001
0.50	A7	5.702923	15.386813	99.859	65.721	62.306	99.865	17.435	18.134	31.255	< 0.001
	A7-OL1	5.829390	27.482684	99.876	68.698	75.732	99.918	93.545	94.541	98.713	< 0.001

4.6 Trained and post-hoc full-pass frontier

Figure 3 places the trained endpoints and uniform TEAL-style clipping of the Run 004 controls on one validation-loss–R_{model} plane. Its A4-OL1 series is the corrected four-site Run 015 result; historical Run 012 A4-OL1[`h`] is excluded from the main frontier and retained in Appendix A. The figure also includes all five A7 and five A7-OL1 full-pass endpoints; Table 11 supplies their matched thresholds and count-pooled site values.

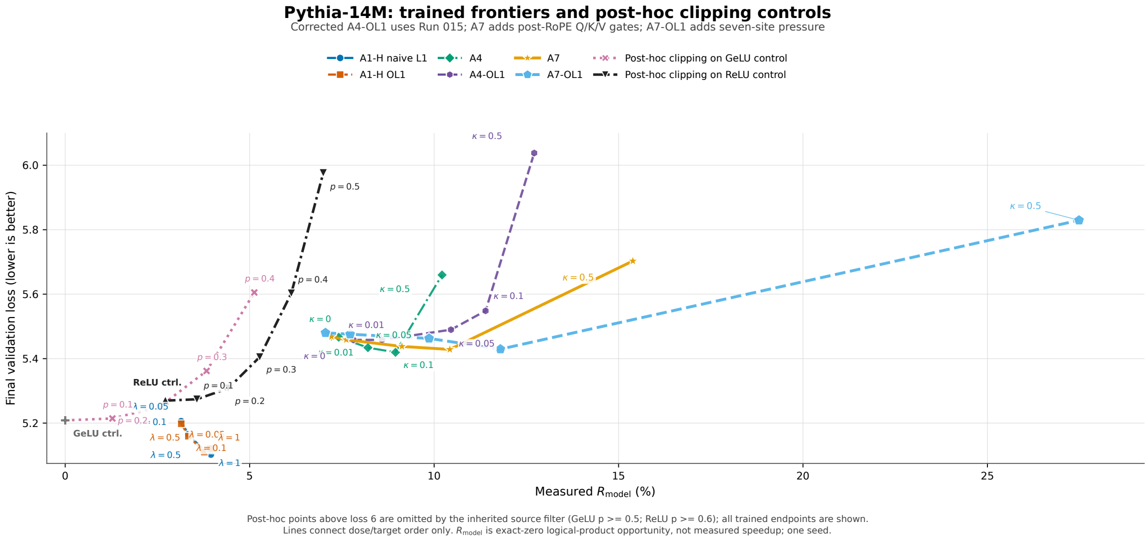


Figure 3: Corrected trained full-pass frontier, uniform TEAL-style post-hoc clipping of the Run 004 controls, Run 015 four-site A4-OL1, and the matched Run 013 A7 and Run 014 A7-OL1 endpoints. A7 uses one-sided thresholds at `a,m,h,z` and symmetric thresholds at post-RoPE `q,k,v`; A7-OL1 adds seven-site orthogonal L1 pressure. All trained endpoints are shown through loss 6.1; post-hoc points above loss 6 remain omitted by the inherited source filter. Lines indicate dose or target order only. Source: Analysis 008, Observation O001.

A Historical A4-OL1[h] realization (Run 012)

Run 012 declared A4-OL1 pressure at `a,m,h,z`, but its frozen training code constructed the pressure capture with `["h"]`; the realized intervention was therefore A4 gates plus OL1 pressure only at `h`. The five endpoints remain valid for that realized A4-OL1[h] intervention, but they are excluded from the main four-site A4-OL1 result and cannot support discarded Finding F001.

Table 12: Historical Run 012 A4-OL1[h] full-pass endpoints. Exact-zero percentages pool integer counts over all six layers and 338 complete validation blocks. Source: Analysis 009, Observation O001.

κ	Loss	R_{model} (%)	Exact-zero (%)						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	5.216	8.409	93.543	49.816	49.413	64.671	< 0.001	< 0.001	< 0.001
0.01	5.205	8.586	94.871	50.390	49.942	71.971	< 0.001	< 0.001	< 0.001
0.05	5.196	9.036	97.915	52.391	51.791	88.307	< 0.001	< 0.001	< 0.001
0.10	5.229	9.343	99.125	54.747	53.809	96.774	< 0.001	< 0.001	< 0.001
0.50	5.723	10.227	99.944	66.441	63.622	99.934	< 0.001	< 0.001	< 0.001

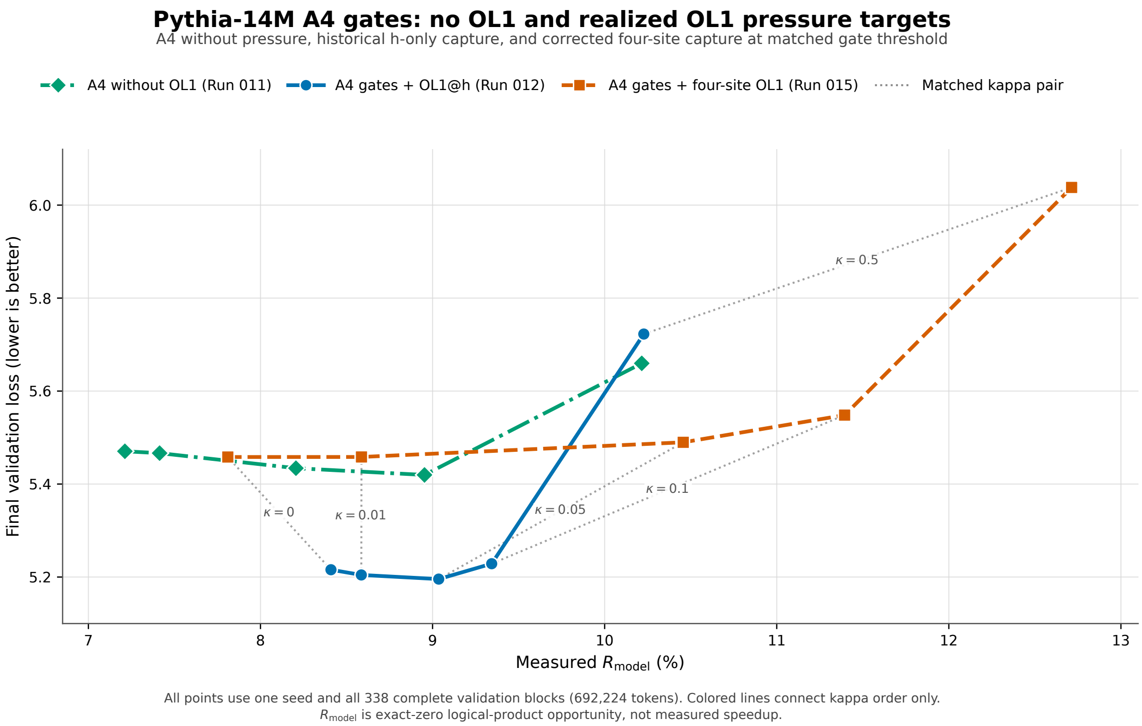


Figure 4: Matched A4, historical Run 012 A4-OL1[h], and corrected Run 015 four-site A4-OL1. The historical and corrected pressure objectives are scientifically different interventions. Source: Analysis 009, Observation O001.

Run 012 used five A100 80 GB Pods for 70–100 minutes per condition and cost \$14.77. Including this historical run, total recorded spend across the six main-ladder runs plus Run 012 is at least \$103.16; the Run 015 billing caveat stated in Section 3 still applies.